

SE3010: Software Engineering Process & Quality Management

Year 3 Semester 1

2026

|                                 |                |
|---------------------------------|----------------|
| Assignment Title                | Assignment 1   |
| Learning outcomes covered       | LO 1:<br>LO 2: |
| Assignment Mode                 | Online         |
| Maximum Marks                   | 100%           |
| Contribution to the Final Grade | 20%            |
| Date published                  | 2026/02/21     |
| Deadline for submissions        | 2026/03/07     |
| Mode of Submission              | Online         |

Assignment description

**Part 1: Tool Selection**

Each group must select **one software testing tool** from the list below:

- PyTest, TestNG, Cucumber, Cypress, Playwright, Robot Framework, Karate, JMeter, Gatling, NUnit, Mocha / Chai, Vitest, RSpec, Behave, k6, Locust

**Selenium, JUnit and Postman are strictly not allowed.**

**Part 2: Each group must identify one industry professional currently working in a software company.**

The industry professional **must have hands-on, practical experience with the same software testing tool selected by the group.**

### Industry Interview – Discussion Areas

The interview discussion should focus on how the selected tool contributes to Quality Engineering practices across the software development lifecycle, rather than on testing alone.

#### A. Background in Quality Engineering

- Briefly describe your role and responsibilities related to Quality Engineering within your organization.
- How long have you been involved in software quality, testing, or quality engineering?
- What types of projects do you primarily work on (e.g., web, mobile, enterprise, cloud, distributed systems)?

#### B. Quality Engineering Tool Experience and Context

- How is the selected tool used to ensure and improve software quality in real projects?
- How long have you been using this tool as part of your quality engineering practices?
- At which stages of the software development lifecycle is this tool mainly used (e.g., design, development, CI/CD, post-deployment)?

Arachana

### C. Quality Engineering Practices Using the Tool

- Which features of the selected tool contribute most to quality assurance and defect prevention?
- What types of quality checks are implemented using this tool (e.g., unit quality, integration quality, system reliability, performance quality)?
- How are test cases or quality checks structured, reviewed, and maintained to ensure long-term quality?

### D. Quality Risks and Automation Challenges

- What quality-related challenges arise when using this tool in real projects?
- How do you handle quality risks such as flaky tests, unstable pipelines, or unreliable test results?
- In which situations is human-driven quality validation preferred over automation, and why?

### E. Quality Engineering in CI/CD and DevOps

- How is the selected tool integrated into the CI/CD pipeline as part of quality gates?
- How frequently are automated quality checks executed, and at what stages?
- How are quality metrics, test results, and failures communicated to developers and stakeholders?

### F. Skills, Expectations, and Professional Advice in Quality Engineering

- What level of quality engineering or test automation expertise do you expect from fresh graduates?
- Which skills should students focus on to become effective Quality Engineers rather than only testers?

- What common quality-related mistakes do junior engineers make when adopting automation tools?
- How does academic teaching of software testing differ from industry-level quality engineering practices?
- What advice would you give students aiming to pursue a career in Quality Engineering or Test Automation?

### Part 3: Presentation Requirements

Each group must deliver a short presentation covering the following aspects:

- Introduction to the selected tool
- Justification for selecting the tool
- Key features of the tool
- Industry adoption and how widely the tool is used

The presentation should be 10 minutes maximum.

### Part 4: Demonstration Requirements

Each group must demonstrate the **complete workflow of using the selected testing tool**, covering the process from **initial setup** to successful **test execution**.

- Each member needs to demonstrate **at least 1 key feature (Per group 4 features)** of the tool, such as:
  - Assertions
  - Fixtures or setup/teardown mechanisms
  - Mocking or stubbing
  - Behaviour-Driven Development (BDD) syntax
  - Test reporting
  - CI/CD usage, etc.

- Demonstration 10 minutes maximum

#### **Part 4: Report**

The report should **demonstrate critical reflection on the insights obtained from the interview discussion.**

- 8-10 pages
- Include workload distribution of the team members
- Page numbers must appear in the **footer**
- Cover Page (Mandatory)  
Must include:
  - Course code and course name
  - Assignment title (Assignment 1)
  - Group number
  - Names and Registration numbers of all group members
  - Submission date

#### **Deliverables**

Each group must submit the following:

- Presentation slides
- Report
- Live demonstration
- LinkedIn profile link of the interviewed industry professional – include in report
- Video interview recording link – Include in report.
  - All **four group members must participate** in the interview
- Plagiarism report - The plagiarism should be less than 10%

**A mandatory viva will be conducted after the presentation and demonstration to assess individual understanding, practical knowledge, and active participation of all group members.**

### Marking Scheme

| Criteria                                                                                                                                                                                | Weight | Excellent (90-100%)                                                                                                                                                                                         | Good (75-89%)                                                                                                                    | Satisfactory (60-74%)                                                    | Poor (40 – 59%)                                                             | Very Poor (0-39%)                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Interview</b><br>Assesses the depth and relevance of insights from the industry professional, focusing on Quality Engineering across the SDLC.                                       | 25%    | Deep, analytical discussion. Strong focus on SDLC integration, CI/CD, automation challenges, quality risks, and professional practices. Clear reflection beyond summary. All members actively participated. | Covers most required discussion areas. Good understanding of tool usage and QE practices. Some analytical reflection is present. | Interview completed, but mostly descriptive. Limited depth and analysis. | Superficial interview. Generic responses. Weak link to Quality Engineering. | Invalid interview, tool mismatch, missing evidence, or minimal engagement. |
| <b>Report</b><br>Assesses the ability to critically reflect on the interview insights and connect them with theoretical concepts from Software Engineering Process & Quality Management | 25%    | Strong critical reflection. Clear linkage between industry insights and Quality Engineering theory. Well-structured, professionally written, and meets all formatting requirements.                         | Good structure and reflection. Minor analytical or formatting weaknesses.                                                        | Mostly descriptive. Limited critical analysis. Some structural issues.   | Weak organization and minimal reflection.                                   | Incomplete or missing required components.                                 |
| <b>Presentation</b><br>Assesses clarity,                                                                                                                                                | 15%    | Professional slides, strong justification,                                                                                                                                                                  | Clear and structured presentation.                                                                                               | Basic explanation.                                                       | Disorganized or unclear explanation.                                        | Incomplete or weak presentation.                                           |

|                                                                                                                                             |     |                                                                                                                                                                                         |                                                                             |                                                      |                                                      |                                                               |
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| structure, justification of tool selection, and communication effectiveness.                                                                |     | accurate explanation of features and industry adoption. Confident delivery.                                                                                                             | Minor clarity or delivery issues.                                           | Limited depth or uneven participation.               |                                                      |                                                               |
| <b>Demonstration</b><br>Assesses practical implementation, correct tool usage, and feature demonstration in supporting Quality Engineering. | 25% | Complete workflow demonstrated. Each member demonstrates at least one key feature (minimum four features per group). Clear explanation of quality benefits. Smooth, reliable execution. | Functional demo with required features implemented. Minor explanation gaps. | Demo works but lacks depth or full feature coverage. | Demo partially works. Limited understanding evident. | Demo is not functional or lacks required features.            |
| <b>Viva</b><br>Assesses each member's individual understanding of the selected tool and related Quality Engineering concepts.               | 10% | Demonstrates strong conceptual and practical understanding. Confident and technically accurate responses.                                                                               | Good understanding with minor gaps.                                         | Basic understanding. Limited depth in responses.     | Weak technical understanding.                        | Unable to explain tools, implementation, or concepts clearly. |